Jordan Watkins, PhD

Quantitative engineer — Bayesian modeling, optimization, PhD mathematics

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PhD mathematician and software engineer who builds probabilistic models end to end. Across several seasons of a competitive league I built and refined a Bayesian paired-comparison rating system — deriving the gradients by hand, writing the optimizer, fitting a heavy-tailed link function and the prior variance to historical match data, calibrating against a known baseline, and modeling correlated rating uncertainty (which meant patching a C++ Bayesian Elo engine) — then scheduled matches with an integer program that maximizes the information each result provides. Ten years architecting and maintaining production Python services alongside it. Published original research in differential geometry. Seeking machine-learning, modeling, or quantitative engineering roles; open to internships and to relocation.

MODELING & QUANTITATIVE WORK

Competitive Rating & Matchmaking Systems · CoNDOR · open source

multiple seasons, through 2022

github.com/incnone/condor_ratings

A Bayesian rating model, series-outcome forecasting, and automated matchmaking, rebuilt and refined across several league seasons (Python, C++, MySQL, PuLP)

- Built and iterated on a Bayesian paired-comparison rating model: per-player Gaussian priors combined with binomial match evidence, fit by MAP estimation
- Derived the log-likelihood gradient analytically and wrote the optimizer by hand — batch gradient ascent with L2 convergence tests on both parameter change and gradient norm, iteration and wall-clock limits, and log-likelihood tracing
- Selected the prior variance by maximizing log-likelihood on held-out matches from a previous season — model selection against real data rather than hand-tuning
- Judged a normal model's tails too thin for real performance variation and designed a heavier-tailed link function — a mixture of an error function and a rational sigmoid whose tail weight is a hyperparameter fit to past match data ($\alpha = 0.42$) — calibrated to agree with the Bradley–Terry model at a chosen reference rating
- Patched Rémi Coulom's C++ Bayesian Elo engine (bayeselo) to expose the covariance matrix it computes internally, then modeled two players' ratings as jointly normal so series predictions account for correlated rating uncertainty
- Converted ratings into calibrated win probabilities and full best-of-3 / best-of-5 outcome distributions, exported as JSON for downstream use
- Automated weekly matchmaking as an integer linear program (PuLP/CBC) maximizing an information-theoretic objective — the entropy of the predicted result, penalized by rating gap — under per-player degree constraints
- Added a local-search pass performing minimum-cost pairwise exchanges to enforce a minimum cycle length in the matchup graph, with unit tests

Architect & Maintainer — Necrobot · Open source

2015–Present

github.com/incnone/necrobot

Discord bots managing competitive speedrunning races and league seasons in *Crypt of the NecroDancer*, in continuous community use for 10+ years

- Architected one codebase that runs two distinct production bots (Necrobot and Condorbot) — ~150 modules across 16 packages with a documented, acyclic package dependency order
- Built an internal command framework separating input parsing and user-facing error handling from data manipulation, over an object layer mapped to MySQL tables
- Async Python (asyncio, aiohttp, discord.py) with MySQL persistence, Google Sheets integration for standings, and timezone-aware match scheduling
- Race ratings built on TrueSkill (Bayesian), adapted to the race format of the game
- Lead architect and maintainer across 10+ years, coordinating a small contributor team; MIT-licensed with admin, code, and database documentation

SKILLS

Modeling	Bayesian inference, MAP estimation, paired-comparison models, heavy-tailed likelihoods, model selection on held-out data, calibration, regularization
Optimization	Analytic gradients, gradient ascent, convergence criteria, integer linear programming (PuLP/CBC), local search
Mathematics	PhD differential geometry; probability, linear algebra, real analysis, information theory
Programming	Python, C++, SQL; async services, data pipelines, long-lived production codebases
Data & tooling	MySQL, pandas, scikit-learn, PyTorch, Google Sheets API, Git

EDUCATION

Ph.D. in Mathematics · University of Michigan — 2013

Specialization: differential geometry (dynamics on Riemannian manifolds). Thesis: "The Rank Rigidity Theorem for Manifolds With No Focal Points."

B.A. in Mathematics and Physics · Knox College — 2007

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PUBLICATION

"The Higher Rank Rigidity Theorem for Manifolds With No Focal Points" · ***Geometriae Dedicata*, vol. 164** — Aug 2012 · pp. 319–349

Springer · arXiv:1111.5428

ENGINEERING EXPERIENCE

Co-Developer — *Bean and Nothingness* · RBOR Games — 2020–2021 · shipped Dec 2021

store.steampowered.com/app/1706090/Bean_and_Nothingness/

Logic-puzzle game · 98% positive from 198 Steam reviews · prototype work from 2012

- Co-lead programmer (C++, SDL, OpenGL): replay/undo state system, GUI level editor shipped via Steam Workshop, custom sound mixer, core game-loop and entity logic
- Lead puzzle designer: created 100+ and edited all ~200 puzzles

CONTINUED EDUCATION IN ML & DATA

Mathematical Physics Seminar · University of Virginia — 2024–Present

Regular participant. Ongoing engagement with current research in geometry and mathematical physics.

Erdős Institute — Data Science Boot Camp · coursework

Applied data science in Python (pandas, scikit-learn).

Self-directed study of machine-learning fundamentals, including building neural networks in PyTorch; ~4,000 reputation and 86 answers on Mathematics Stack Exchange.

Career note: independent study and freelance tutoring 2013–2020, during part of which I carried family caregiving responsibilities.